

Denis Bolonin

Senior Software Engineer, 5+ years of experience

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UK full-time work permit, valid until Jan 2028

London, UK

PROFESSIONAL SUMMARY

Senior Software Engineer with **over 5 years** of commercial experience, working in teams ranging from small startups to large fintech companies, both remote and on-site. Built full-cycle development processes, from gathering requirements to establishing fully automated release pipelines with test coverage and CI/CD. Expert in developing high-performance backend systems and integrating various technologies.

Being Agile certified, I'm involved in projects from start to finish - defining tasks, working with analysts' team and even customers, developing solutions, and managing releases. My team lead granted me the chance to take on some leadership responsibilities, so I had a positive experience mentoring new colleagues who were just starting out on our team.

Actively engaged in the open-source community, I've contributed to projects like ClickHouse, a prominent open-source DBMS, with technical work recognized by the project's CTO. Recent highlights include an over 80% speed-up in contract-notification processing and financial model tooling for a 2 billion USD investment project.

WORK EXPERIENCE

Senior Software Engineer

08/2024 - Present

[digital.sfpe](#), a software development SaaS startup

- Developed from the ground up a comprehensive software solution tailored for fire protection engineering companies to ensure regulatory compliance. This solution automates critical processes such as contract management, data validation, format control, minimizes errors and proactively monitors deadlines. The implementation of this software significantly improved efficiency, enabling a **single person** to manage the workload that previously required **3-4 full-time employees**.
- Implemented autofill process for EMERCOM contract notifications, reducing the processing time per notification from 8-10 minutes to under 1.5-2 minutes (**over 80% speed up**).
- Engineered an end-to-end subscription billing system (order intake, payment document parsing, invoicing, and reconciliation), replacing an almost fully manual workflow and eliminating human error across the billing pipeline.

Software Developer

07/2022 - 07/2024, 2 years 1 month

[Sber](#), one of the biggest fintech companies

- Developed a utility for financial models for a high-value investment project, specifically valued at 2 billion USD, **accelerating** model calculations **by over 50%** through concurrent processing.
- Developed an automated data pipeline for depositary settlements, enabling fully automated data streams and eliminating manual intervention.
- Developed a team productivity dashboard for Scrum Masters, enabling data-driven decision-making and **increasing team throughput by 15%**.

Open Source Developer

10/2020 - 06/2022, 1 year 9 months

[ClickHouse](#), opensource DBMS

- Researched and implemented UDP-based replication protocols (UDT, μ TP/uTP, ENet) for ClickHouse, evaluating their performance and laying groundwork for future QUIC protocol studies.
- Integrated Rust into ClickHouse build system, enabling C++ and Rust interoperability. Implemented and integrated the BLAKE3 hashing function, achieving almost **100% performance increase** with a new hashing function compared to existing functions.
- Implemented YAML configuration support for ClickHouse, providing users with a more readable and convenient configuration format.

TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript, C, Rust, SQL, Bash, R

Backend Development: Django, FastAPI, Flask, REST APIs, Gunicorn, Nginx

Web Scraping & Data Extraction: BeautifulSoup, Selenium

Frontend Development: jQuery, React

Databases & Data Storage: PostgreSQL, SQLite, ClickHouse, Redis

Cloud & DevOps: Docker, Kubernetes, OpenShift, Jenkins, Helm, AWS, Istio, Nexus, CI/CD

Monitoring & Analytics: Prometheus, Grafana, Kibana

Message Queues & Streaming: Apache Kafka, RabbitMQ, ZooKeeper, Apache Spark

Development Tools: VS Code, PyCharm, Jupyter Notebook, Git, Conda, CMake, Linux, Bitbucket, GitHub, GitLab

Project Management: Agile, Scrum, Jira, Confluence, Team Leadership, Mentoring, Figma

Testing: pytest, Fuzzing, Unit Testing, Functional Testing

EDUCATION

Master's degree in Computer Science. Queen Mary University of London, London, UK 09/2024 - 09/2025

Software and Data Engineering stream

Bachelor's degree in Computer Science. HSE University, Moscow, Russia 09/2019 - 06/2023

Applied Mathematics and Informatics, Distributed systems

CERTIFICATIONS

ICAgile Certified Professional 12/2022

Completed Agile training, received the ICAgile Certified Professional

[Certificate Link](#)

Applied statistical analysis 05/2022

Completed additional Applied Statistical Analysis course, HSE University

PUBLICATIONS

More Than 2x Faster Hashing in ClickHouse Using Rust

Co-authored technical blog post on integrating Rust into ClickHouse. The work was recognized by ClickHouse CTO Alexey Milovidov, who wrote: "I'm highly impressed by the work of Denis Bolonin that made it possible!"

[ClickHouse Official Blog](#)

ACHIEVEMENTS

Winner of the Finance block hackathon at Sber 09/2023

- Developed the backend data layer for a mobile approval platform, including storage and management of user data and incoming approval requests.
- Collaborated with the team on product ideation and workflow design.
- The solution replaced a cumbersome legacy process with a more engaging, gamified approval experience.